

PEARLAND, TX, USA

WMO: 722427

Lat: 29.519N

Lon: 95.242W

Elev: 13

StdP: 101.16

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 12975

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.5	2.4	-8.6	1.8	7.3	-6.2	2.2	8.8	10.0	14.0	9.0	13.9	2.5	350	0.392

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.8	35.3	25.0	34.4	25.0	33.6	24.9	26.8	31.2	26.4	30.9	26.1	30.6	3.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.9	21.2	28.4	25.3	20.4	28.2	25.0	20.1	28.1	83.6	30.8	82.0	30.6	80.7	30.7	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.5	7.6	7.0	DB	-2.2	37.7	2.2	1.2	-3.8	38.6	-5.1	39.3	-6.4	40.0	-8.0	40.9	
			WB	-4.0	27.6	1.9	0.8	-5.4	28.2	-6.4	28.6	-7.5	29.1	-8.8	29.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.6	12.5	14.6	18.0	21.3	25.1	28.2	28.9	29.2	27.3	22.8	17.4	13.9
	DBStd	6.97	5.10	5.19	4.40	3.42	2.58	1.49	1.42	1.55	1.96	3.85	4.81	5.10
	HDD10.0	74	31	16	3	0	0	0	0	0	0	0	4	19
	HDD18.3	629	191	124	60	14	1	0	0	0	0	10	73	156
	CDD10.0	4318	109	144	252	339	468	545	584	594	518	396	227	141
	CDD18.3	1829	10	19	51	102	211	295	326	336	268	147	46	19
	CDH23.3	18368	26	66	201	619	1903	3377	3838	4005	2707	1284	271	71
CDH26.7	7373	0	7	21	97	586	1496	1749	1881	1099	397	38	3	
Wind	WSAvg	2.9	3.4	3.6	3.6	3.6	3.4	2.6	2.0	2.0	2.2	2.6	2.9	3.3
Precipitation	PrecAvg	1352	108	76	94	103	132	148	115	109	119	136	107	105
	PrecMax	2020	267	231	215	250	335	625	308	231	291	446	322	210
	PrecMin	602	15	13	20	2	1	7	35	13	41	1	18	28
	PrecStd	331	60	56	54	77	84	121	73	66	59	114	70	43
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.7	27.7	28.7	30.4	33.1	36.2	36.1	36.9	35.2	32.9	29.7	26.9
		MCWB	18.6	19.1	20.3	21.2	23.2	23.7	24.7	24.9	24.1	24.0	22.7	21.3
	2%	DB	23.9	25.2	26.9	28.9	31.9	34.2	34.9	35.6	33.9	31.6	27.7	25.2
		MCWB	18.8	19.7	20.1	21.2	23.1	24.5	25.1	25.2	24.4	23.9	21.9	21.1
	5%	DB	22.5	23.6	25.6	27.7	30.8	33.2	34.0	34.5	32.9	30.3	26.3	23.8
		MCWB	18.6	19.4	19.8	21.3	23.1	24.5	25.1	25.3	24.3	23.2	21.1	20.5
	10%	DB	21.1	22.3	24.1	26.7	29.8	32.4	33.1	33.5	32.0	28.9	24.8	22.4
		MCWB	18.0	19.0	19.4	20.8	22.9	24.5	25.1	25.2	24.1	22.2	20.7	19.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.5	22.4	22.8	24.7	25.8	27.2	27.0	27.3	26.9	26.3	24.7	23.2
		MCDB	23.2	24.1	25.2	27.8	29.6	30.9	31.4	32.1	30.5	29.1	27.1	24.9
	2%	WB	20.7	21.6	22.0	23.8	25.3	26.6	26.6	26.8	26.3	25.5	23.6	22.3
		MCDB	22.5	23.6	24.5	26.7	28.7	30.4	31.3	31.6	30.2	28.9	26.1	24.2
	5%	WB	19.8	20.8	21.5	23.0	24.7	26.0	26.2	26.4	25.8	24.9	22.7	21.4
		MCDB	21.8	22.7	23.9	25.7	28.2	30.1	31.0	31.3	29.7	28.4	25.2	23.4
	10%	WB	18.6	19.8	20.9	22.4	24.3	25.6	25.8	26.0	25.4	24.3	21.7	20.2
		MCDB	20.7	21.8	23.2	25.1	27.8	29.9	30.5	30.9	29.4	27.7	24.2	22.1
Mean Daily Temperature Range	5% DB	MDBR	10.1	9.5	9.4	9.3	8.6	8.6	8.5	8.8	8.7	10.2	10.5	9.7
		MCDBR	10.9	9.8	9.4	9.3	9.5	9.8	9.7	10.1	9.8	9.8	10.3	10.0
		MCWBR	7.3	5.9	4.8	4.3	3.4	2.9	2.7	2.7	3.0	4.0	5.5	6.8
	5% WB	MCDBR	8.9	8.0	7.6	6.9	7.4	7.8	8.5	9.0	8.0	8.2	8.6	9.1
		MCWBR	7.4	6.2	4.9	4.1	3.2	2.9	2.8	2.8	3.1	4.2	5.8	7.2
Clear-Sky Solar Irradiance	taub	0.344	0.361	0.375	0.439	0.480	0.471	0.499	0.485	0.457	0.380	0.372	0.353	
	taud	2.505	2.466	2.416	2.236	2.157	2.235	2.178	2.217	2.263	2.471	2.441	2.484	
	Ebn at Noon	892	905	912	860	822	825	801	809	821	873	850	861	
	Edh at Noon	90	102	114	141	153	141	149	142	131	100	96	89	
All-Sky Solar Radiation	RadAvg	2.89	3.26	4.47	5.32	5.95	6.16	5.94	5.71	4.96	4.36	3.30	2.57	
	RadStd	0.28	0.55	0.38	0.39	0.42	0.49	0.57	0.48	0.53	0.52	0.33	0.30	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (34 neighbors)	+0.33	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+111	+97

Nomenclature: See separate page